Day 2: Diagnosing Hardware Failures Theory (30 min):Common hardware issues and their symptoms.Practical (30 min): Test components using a multimeter.

1.List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' symptom: blank screen, no POST beep).

ANS:

No.	Hardware Issue	Typical Symptom
1	Faulty RAM	Computer fails to boot, shows a blank screen, or produces beep codes.
2	Power Supply Unit (PSU) Failure	Computer does not turn on, or it shuts down unexpectedly.
3	Hard Drive or SSD Failure	Computer boots slowly, displays disk errors, or fails to load the operating system.
4	Overheating CPU	Computer becomes very hot, runs slowly, or shuts down unexpectedly.
5	Faulty Graphics Card	No display, distorted graphics, or colored lines on the monitor.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

ANS:

Laptop Not Charging – Step-by-Step Troubleshooting

Step 1: Check the power socket.
Plug another device into the socket to check whether electricity is available.

Step 2: Check the charger.
Look at the charger and cable. Check for cuts, broken wires, or damage.

Step 3: Connect the charger properly.
Make sure the charger is connected properly to the laptop and the power socket.

Step 4: Check the charging light.
Look at the laptop to see whether the charging light is on.

Step 5: Check the charging port.
Look for dust or damage in the charging port. Do not put metal objects inside it.

Step 6: Restart the laptop.
Turn off the laptop, unplug the charger, and hold the power button for 15 seconds. Then reconnect the charger and check again.

Step 7: Check the charger voltage.
Use a multimeter to check the charger's output voltage if you know how to use it safely.

Step 8: Try another charger.
Use a compatible working charger to check whether the original charger is faulty.

Step 9: Check the battery.
Look for swelling or damage. If the battery is swollen, stop using the laptop.

Step 10: Visit a technician.
If the laptop still does not charge, the battery, charging port, or motherboard may have a problem.

3.Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

ANS:

Step 1: Take a multimeter and set it to DC voltage (V⎓).

Step 2: Connect the phone charger to a power socket.

Step 3: Carefully place the multimeter probes on the USB charger's output using a safe test adapter.

Step 4: Read and record the voltage shown on the multimeter.

Step 5: Compare the measured voltage with the expected range of 4.75 V to 5.25 V.

Observation Table

Item	**Result**
Device tested	USB phone charger
Expected voltage	4.75 V – 5.25 V
Measured voltage	5.02 V (example reading)
Result	Within the expected range
Conclusion	The charger output voltage is normal in this example.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

ANS:

Step-by-step procedure

Step 1: Turn off the PC and unplug the power cable from the wall socket.

Step 2: Disconnect the ATX power connector from the motherboard.

Step 3: Set the multimeter to DC voltage (V⎓).

Step 4: Identify the wires. The yellow wire is +12V and the black wire is ground (0V).

Step 5: For a voltage measurement, the power supply must be running. Use a proper PSU test adapter and follow the manufacturer's safe testing procedure. Do not probe a loose connector or short pins while the PSU is energized.

Step 6: Place the red probe on the yellow wire's +12V output and the black probe on a black ground wire.

Step 7: Read the voltage displayed on the multimeter and record the result.

Observation table

Item	Result
Expected 12V rail voltage	Approximately 12V
Acceptable ATX range	11.40V–12.60V
Example measured voltage	11.30V
Result	Below the expected range
Possible problem	Faulty or weak power supply (PSU)